

BENJAMIN E. JORDAN

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LinkedIn | Personal Website | GitHub

SKILLS

Languages: Python, Java, Scala, TypeScript, C++ **Tools:** Git, Docker, AWS, Runpod, Linux, HuggingFace, LaTeX

Libraries: PyTorch, NumPy, Wandb, PySpark, Pandas, vLLM, Boto3, Faiss, Optuna, OmegaConf, Pytest

EXPERIENCE

Software Engineer @ Amazon Oct 2024 – Present
Measurement, Ad Tech, & Data Science New York, NY

- Productionized and owned team's brand-awareness model inference system (based on AWS SageMaker), which processes billions of rows of data daily, aggregating survey responses and model predictions for brand lift reports
- Designed and implemented a workflow to automate model training, evaluation, and approval on a weekly cadence
- Designed an online model shadowing system that enabled faster model testing and releases
- Designed and implemented a daily feature pre-processing job + data lakehouse, reducing batch inference costs by over \$100k per year and enabling the system to scale cheaply to new marketplaces/regions
- ML system design was referenced by multiple high-visibility projects across the org, presented design to partner teams
- Led the design and development of a reporting feature which automatically compiles study results into presentation-ready PowerPoints, saving hours of manual effort for thousands of users with active studies

Machine Learning Engineer @ Northrop Grumman Feb 2024 – Oct 2024
Autonomous Intelligence & Robotics Lab Denver, CO

- Enabled accurate, real-time object detection by distilling GroundingDINO into YOLOv10, decreasing inference latency on Jetson Orin from 0.6s to 0.08s
- Wrote a report on techniques for mitigating a lack of labeled training data: active learning, semi-supervised learning, knowledge distillation, and synthetic data generation, informing the team's adoption of distillation as a tool for efficiency
- Led the technical development of a project focused on classified PDF retrieval and question answering, and wrote a report covering techniques for implementing and evaluating retrieval-augmented generation (RAG) systems
- Implemented a RAG system utilizing ANN search and locally running language and dense retrieval models, evaluated retrieval performance on dataset of LLM generated Q&A pairs based on documents using MRR and recall@k

Machine Learning Engineer @ KLA May 2023 – Aug 2023
Internship Ann Arbor, MI

- Researched and implemented algorithms for semiconductor defect detection and classification
- Fine-tuned and quantized vision transformers with limited data, outperforming statistical methods while staying efficient
- Presented project during poster board session and was invited to give a virtual talk on transformers and transfer learning

PROJECTS

Entropy Audio | Website | Project Blog Jan 2024 – Present
Personal project focused on novel sound design tools for composers

- Curated a multi-terabyte audio sample dataset using open-source data, web-scraping, synthetic data, and manual labeling
- Used local LLMs (Gemma 4 31B), an RTX 5090, and vLLM to efficiently generate captions for millions of samples
- Performed continued training on a 1.4B diffusion transformer, updated with new text encoders and augmentation (with in-memory embedding caching), implicit-duration conditioning, improved attention masking, and novel offline eval metrics
- Implemented policy gradient reinforcement learning algorithms for diffusion models, using pretrained reward models to better optimize for downstream objectives
- Designed a frontend that incorporates a preference data flywheel into the workflow, and created a backend service using AWS and RunPod to support generation/inference as well as other application logic
- Implemented the loss from the Diffusion-DPO paper and trained a model on a self-collected preference dataset

Open Source Contributions | GitHub April 2026 – Present

- Minor contributions to Google's JAX, a library for high-performance numerical computing and large-scale machine learning

EDUCATION

Cornell University 2022 – 2023
Master's in Computer Science, Concentration in Machine Learning Ithaca, NY

Rochester Institute of Technology 2018 – 2022
Bachelor's in Computer Science, Minor in Music, Presidential Merit Scholarship, Magna Cum Laude Rochester, NY

NOTABLE COURSEWORK

Machine Learning: Large-Scale Machine Learning, Machine Learning Theory, Reinforcement Learning, Computer Vision, Numerical Linear Algebra, Natural Language Processing